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Update the constraints on the long-range new force with the Juno data

Contents

The gas giant Jupiter is the largest planet in our solar system and fifth one from the Sun. It has been observed extensively for much of human history: with a total of eleven space missions sent from Earth to Jupiter so far, Jupiter has been an important topic of study to learn more about planetary science and other branches of astrophysics. The most recent Jupiter mission, the spacecraft Juno [1], was launched in 2011 and has since orbited Jupiter more than 60 times, gathering extensive data with which to study the planet.

In addition to learning directly the properties of Jupiter, recent work by my group [2–4] has shown the unexpected utility of Juno's data for probing physics beyond the standard model of particle physics (BSM).[1] In this note, I propose a pretty straightforward followup to [4] on using the Jupiter's gravitational field to probe long-range new forces.

The basic idea of [4] could be summarized as follows. There has been precise determination of Jovian gravitational field from the orbital motion of the Juno spacecraft. Thus, deviations of Juno's orbit from the expected behavior could be a response to additional forces acting between Juno and Jupiter. Therefore, the data of Juno's orbits, and the degree to which they are explained by Jupiter's gravitational influence, can be a powerful probe of an interaction arising from physics beyond the Standard Model of particle physics, sometimes referred to as a "fifth force."

What has been done is Ref [4] is to use the harmonic expansion of Jupiter's gravitational field [11], which describes the deviation of Jupiter's gravitational potential from spherical symmetry and accounts for modeled astrophysical effects in the vicinity of the Juno-Jupiter system. The uncertainty in these gravitational parameters is used to quantify the uncertainty in the precession of Juno's orbit. Then, to provide constraints on the strength and range of the fifth force, we require that additional precession induced by the fifth force does not exceed the quantified uncertainty inferred using the Jovian gravitational data. Juno's orbit is highly eccentric ($1 - e \sim 0.01$), and its perijoves or points of closest approach to Jupiter are on the order of $0.05\, R_{♃}$, where $R_{♃}$ is the radius of Jupiter, as represented in fig. 1. For this reason, Juno's orbits should be sensitive to deviations arising from a fifth force, particularly in a regime of length scales corresponding to Juno's orbit.

What I propose you to try is as follows. As mentioned above, in Ref [4], one crucial step is that we use the uncertainties in the gravitational harmonic expansion reported in [11] to compute the uncertainty in Juno's precession. Yet in a more recent paper [12], the authors reported a more precise determination of the gravitational harmonic expansion parameters. In particular, for harmonic parameters J_5 to J_{10} which we included in our previous computation, the uncertainites have been improved by a factor of 2 to 20. This might lead to a stronger constraint on the fifth force, which is the main target I want you to check.

[1] See [5–10] for the use and proposed use of other Jupiter data (not necessarily the Juno data though) to study new physics.

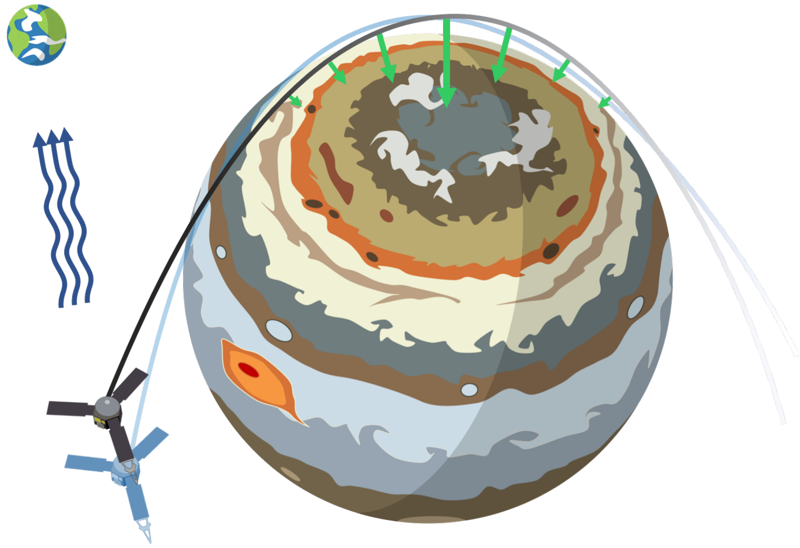

Figure 1. A schematic which highlights the highly eccentric orbit of Juno around Jupiter, as well as its very near approach to Jupiter's surface. Here, the green arrows and their lengths stand for the fifth force that Juno would experience along its trajectory. When the range of the fifth force is small, the fifth force effect is only significant near Juno's closest approach to Jupiter (or perijove) and changes drastically with time. Consequently, Juno's orbit may deviate from our expectations, as shown in the light blue shaded trajectory.

Suggested steps

- As a first step, go through in detail Ref. [4] and reproduce the key results yourselves. I want you to understand the physics target (the fifth force), observable (precession of the Juno spacecraft) and the method (instead of using the trajectory data directly, we use J's to estimate the precession). Though the models of the fifth force are usually constructed as quantum field theories (QFTs) which you haven't learned, in our study we simply parametrize it using a Yukawa potential which doesn't require you to know QFTs. You could work together and set up an analysis pipeline to reproduce some key results, ideally the black solid curve in the final money plot, Fig. 4 in [4].

  Praniti and Shi had set up the codes already. Yet for the training purpose, I want you to develop your own analysis framework for computing the uncertainty of the precession from the Juno harmonic expansion parameters, and precession predicted from fifth force.

- Use the new uncertainties in the gravitational harmonic expansion reported in [12] to test how the constraint on fifth force would change. We don't intend to reproduce [12] (which is a very technical planetary science paper beyond our scope. In addition, their main focus is Jovian wind which we don't care about, while constructing the gravity field is only an intermediate step for them). Yet we want to understand as best as we could the differences in determining the gravitational field harmonics between [12], and [11] which was used in our previous work[4]. As far as I could tell, they both use Juno Doppler data and the most obvious difference is the number of orbits used. It would be useful to read through them to understand a bit better.

Note that [12] (and the database associated with it online) doesn't give the full covariant matrix for the harmonic expansion parameters. Please check and confirm (searching for relevant information is a basic skill for research). If so, I will contact the authors to see if they could share it. If we could find the full covariant matrix (either find it online or through the authors), we implement a full analysis. In the worst case that we could not obtain the new full covariant matrix, we could just change the diagonal entries of the covariant matrix which are provided, as a crude estimate on whether improvements on the fifth-force constraints are possible.

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